01 / EDA EXPLAINED

What this EDA phase did

EDA means Exploratory Data Analysis: inspecting and summarizing data to understand its contents, patterns, and possible problems before cleaning it and building models.

Main outcome: we understand the ticket records and have identified the decisions needed for preprocessing.

The dataset we explored

The notebook loaded railway.csv: 31,653 ticket transaction records and 18 columns. Each record connects a ticket purchase to a planned train journey. It does not identify a unique passenger, verify boarding, or identify a distinct train service.

The columns describe purchases, payments, Railcards, ticket types and classes, prices, stations, journey dates and times, service status, disruption reasons, and refund requests.

Purchase dates span 8 December 2023–30 April 2024. Journey dates span 1 January–30 April 2024. These are different date windows, not contradictory facts.

How the notebook is set up

pandas reads and groups the data. NumPy supports numerical operations. Seaborn draws the statistical charts, while Matplotlib manages figures, labels, and layouts. Seaborn is built on Matplotlib [2]. Its whitegrid theme changes chart appearance, not results.

The simple file path assumes the kernel runs from System/EDA_Phase. The loader preserves literal text such as Railcard "None". The data dictionary is an external reference; it is not loaded as a calculation input.

How to read the final notebook

The notebook contains 24 code cells and 10 saved chart outputs. It is organized by topic: calculations and tables appear before their separate visualization cells. Headings and essential interpretation notes remain, while unnecessary comments and editing history were removed.

Scope of this guide: figures and results come from EDA_final.ipynb [1]. This guide did not independently rerun the source CSV.

02 / EDA EXPLAINED

Missing values and ticket prices

Not every blank is a data error

Actual Arrival Time is empty for 1,880 cancelled records. Reason for Delay is empty for 27,481 on-time records. The notebook checks these patterns by status and retains them; it does not invent an arrival or a disruption reason.

Railcard is different: 20,918 records contain the literal value "None", meaning no Railcard, not missing information. The remaining categories are Adult, Disabled, and Senior. No empty Railcard cells appear in the saved results.

What the price summary tells us

Prices range from 1 to 267 price units. The mean (average) is 23.44; the median (middle value) is 11. The middle half of prices lies between Q1 = 5 and Q3 = 35. These numbers describe the recorded prices, not future prices.

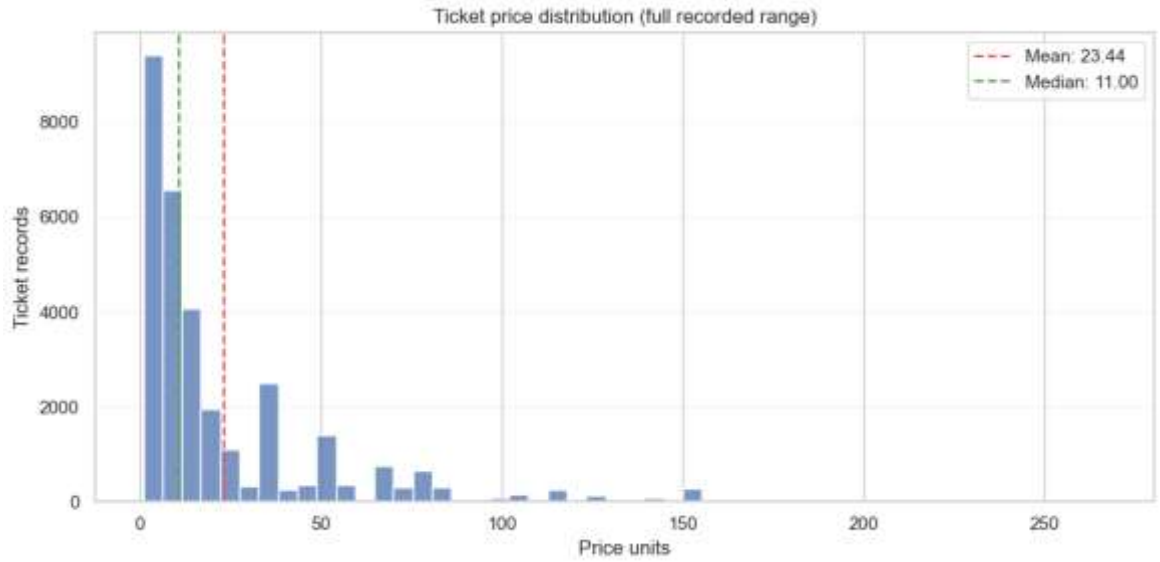


Figure 1. Most ticket records have lower prices; a smaller number form a long high-price tail. The full recorded range is shown.

The histogram counts tickets in price intervals. The mean is above the median because higher prices pull the average upward. This is called a right-skewed distribution. Expensive tickets are not automatically mistakes and are not deleted here.

Ticket sales value = the sum of recorded prices before refund adjustments. The total is 741,921 price units; it is not net revenue or profit.

The CSV does not state a currency. GBP is a project assumption, so the notebook charts correctly use neutral price-unit labels.

03 / EDA EXPLAINED

Ticket groups, purchases, and routes

Compare ticket groups fairly

The notebook groups records by Ticket Type and Ticket Class, then reports counts, usable prices, means, and medians. It plots average prices for each combination. Anytime First Class has the highest group mean (77.23); Advance Standard has the lowest (15.34).

Railcard categories are compared separately using counts, mean prices, median prices, and percentage shares. These are observed differences; route and ticket mix may also differ, so the comparisons do not prove a causal discount effect.

How tickets were purchased

The channel and payment tables show counts and percentages. Online has 18,521 records; Station has 13,132. Credit Card is the most common payment method. A cross-tabulation is simply a table that compares two categories: here, purchase channel and payment method.

Payment percentages are calculated within each channel. For example, Credit Card accounts for 62.0% of online records and 58.2% of station records. The denominator is the number of records in that channel.

Which routes contribute the most sales value?

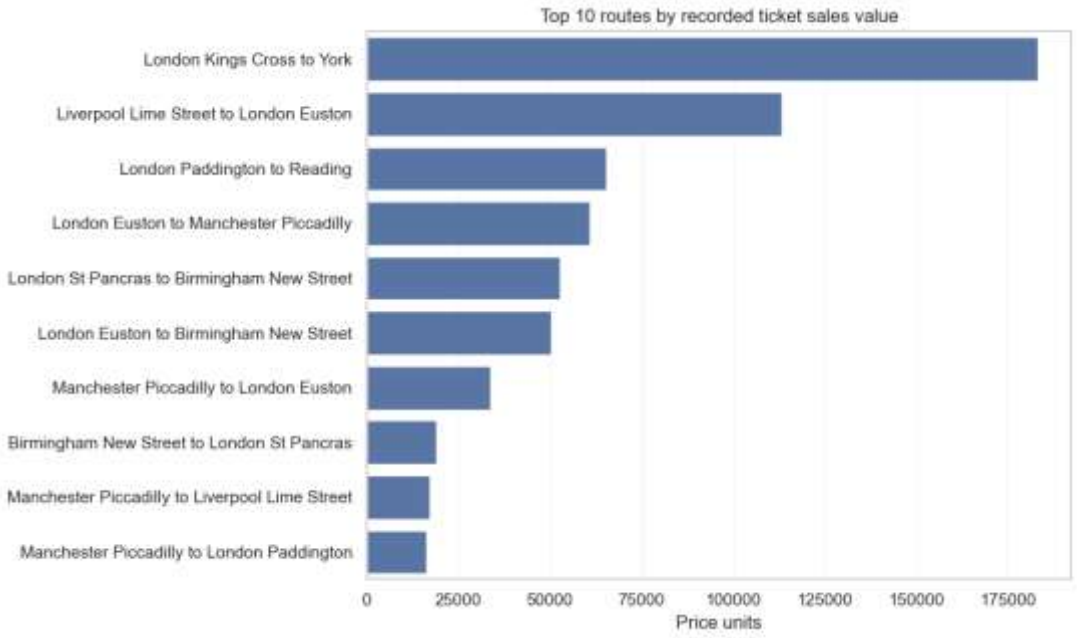


Figure 2. The ten routes with the highest recorded ticket-sales value. Each route is directional: A to B is separate from B to A.

London Kings Cross to York leads with 183,193 price units from 3,922 ticket records. Route tables also include record counts and mean prices. High sales value is not the same as highest ride demand or highest profit.

04 / EDA EXPLAINED

Purchase activity versus travel demand

Purchase date answers: "When was the ticket bought?" Journey date answers: "When was the trip scheduled?" A ticket bought in January for February contributes to January purchases and February journey demand.

Daily and monthly purchase summaries

For each purchase day, the notebook counts records and sums usable prices. It creates a complete date calendar. The 17 dates with no records have zero recorded sales value; dates containing records but no usable prices would remain unavailable, not zero.

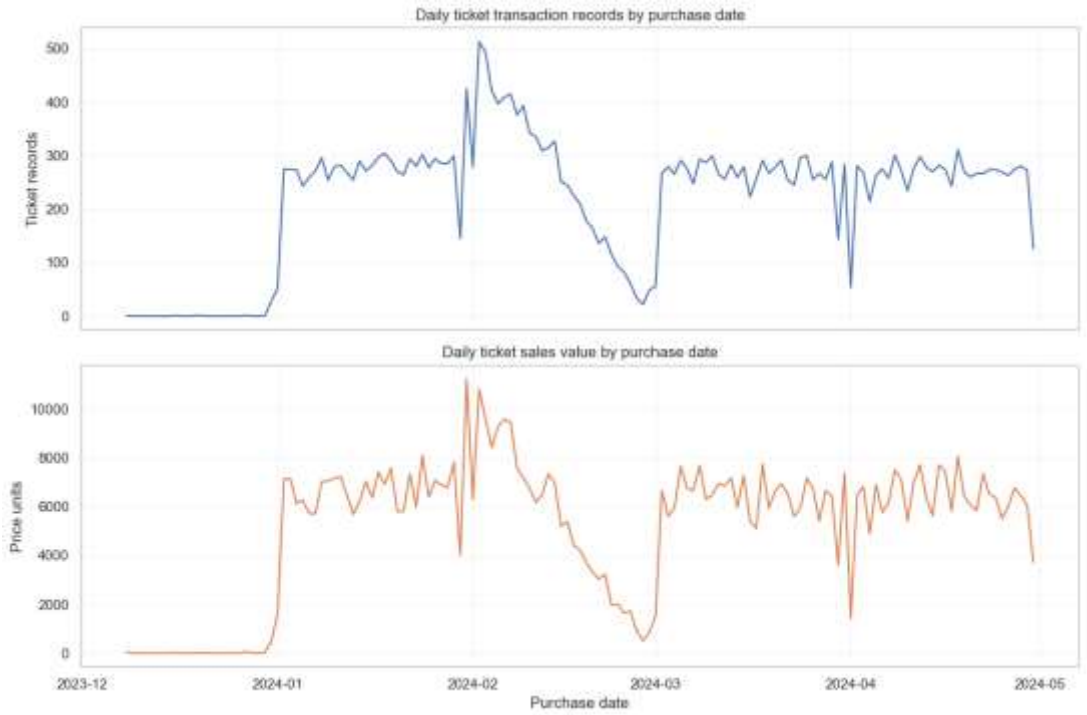


Figure 3. Daily purchase records and ticket-sales value use the same calendar, but have different units.

The same daily measures are added by calendar month. January has the largest monthly purchase total: 8,434 records and 205,091 price units. December contains only 34 records and should not be treated as a normal full month.

Journey-date summaries

Journey-date record totals are 8,111 in January, 7,644 in February, 8,117 in March, and 7,781 in April. The notebook also plots daily journey-date counts. These include cancelled records and are not verified completed rides.

These charts show observed variation, not a forecast. Four months cannot establish annual seasonality, and the extract-selection rule is unknown; purchase-date completeness must not be assumed.

05 / EDA EXPLAINED

Reliability, refunds, and review flags

Recorded journey outcomes

The notebook reports 27,481 On Time records (86.82%), 2,292 Delayed (7.24%), and 1,880 Cancelled (5.94%). These are ticket-record shares, not percentages of distinct train services.

Refund requests: compare percentages, not just counts

Refund tables contain both counts and within-status percentages. There are 546 requests among Delayed records and 572 among Cancelled records. On Time records have no requests in this extract.

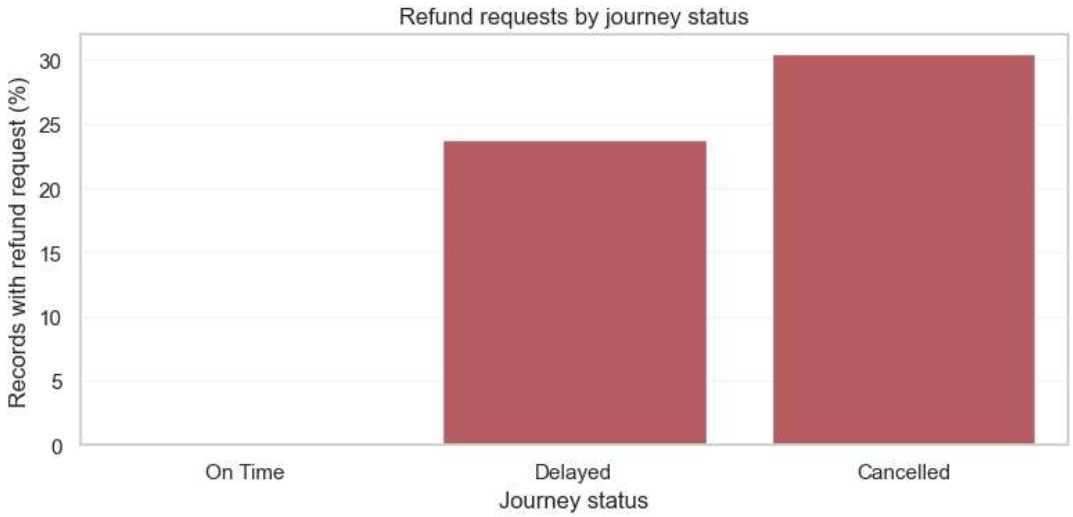


Figure 4. Request rates are 0.00% for On Time, 23.82% for Delayed, and 30.43% for Cancelled ticket records.

For example, 546 ÷ 2,292 × 100 = 23.82%. Using the records in each status as the denominator allows groups of different sizes to be compared. A request does not confirm approval or payment, so its ticket price is not automatically deducted.

Two issues to review during preprocessing

Disruption reasons are counted for Delayed and Cancelled records. Labels such as "Signal Failure" and "Signal failure" vary in capitalization. "Staffing"/"Staff Shortage" and "Weather"/"Weather Conditions" need a meaning-based decision before merging.

To calculate arrival delay, the notebook combines journey date with clock times. If an arrival clock time is earlier than departure, it assumes arrival was the next day. Under this rule, 18 records marked Delayed show arrival at or before schedule. They are retained as review flags, not relabeled.

06 / EDA EXPLAINED

Validation and the next phase

What the final checks confirm

The saved results show 31,653 unique transaction IDs and no duplicate full rows. Purchase and journey dates parse successfully. All 31,653 recorded prices are usable in the summaries.

Record totals agree across the route summary, daily and monthly purchase summaries, journey-date summary, and category counts. Overall, route, daily, and monthly price totals all equal 741,921. This agreement is called reconciliation: different summaries add back to the same source total.

The uploaded notebook has consecutive execution numbers for its 24 code cells, 10 saved chart outputs, and no saved execution errors or warnings. Those are useful checks, but matching totals alone cannot prove that every source value is correct.

What EDA did—and did not do

EDA loaded the data, described it, compared groups, visualized patterns, checked missingness and duplicates, and flagged issues. It did not overwrite the raw CSV, produce a cleaned export, train a forecasting model, or build the Tableau dashboard. It is a completed initial exploration, not the finished four-week project.

Move into Data Preprocessing

Use a separate notebook and work on a copy of the raw data. Convert and validate data types; standardize clear label variants; preserve meaningful missing values and Railcard "None"; and document decisions about the 18 status/time disagreements. Validate date and overnight assumptions before relying on derived delay values.

Create useful fields such as route, booking lead time, weekday, departure hour, and a review flag where appropriate. Then export the cleaned dataset and recheck record counts and price totals. No rows or expensive prices should be removed without a documented reason.

Ready to proceed: keep EDA_final.ipynb as the initial EDA record, begin preprocessing, then briefly recheck the analysis after cleaning.

Sources

[1] EDA_final.ipynb — supplied project notebook. All dataset figures and chart images in this guide use its saved results; the source CSV was not independently rerun for this guide.

[2] Seaborn official documentation — “seaborn: statistical data visualization”. https://seaborn.pydata.org/ — accessed 31 October 2026. Used only for the library description.